National University of Science and Technology

Atta-ur-Rahman School of Science and Technology

Project Report

INDIGENOUS DEVELOPMENT OF DNA TAGGING SYSTEM FOR INDUSTRIES

May, 2026

Project Approvals

This written technical brief documents the system architecture, operational workflow, mathematical calculations, and empirical qPCR verification of the **DNA Base Tag and Trace System**.

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Chapter 1: Project Background & Objectives

1.1 National Strategic Context

Illicit trade, product adulteration, counterfeiting, and tax stamp evasion impose severe risks across sovereign supply chains, public healthcare systems, and national revenues. In developing and transitional economies, critical supply chains suffer from systemic vulnerabilities:

1. **Pharmaceuticals and Therapeutics**: Counterfeit medicines and sub-potent formulations lead to treatment failure, antimicrobial resistance, and loss of life.
2. **Petroleum and Fuels**: Adulteration of refined petroleum with substandard solvents causes industrial equipment failure and massive customs revenue losses.
3. **Defense and Strategic Materials**: Component tampering, unauthorized diversion, and unverified batch substitution compromise operational safety and national security.
4. **Excise and Tax Stamps**: Document forgery and cloned fiscal stamps bypass government excise collection on alcohol, tobacco, and high-value manufactured goods.

To counteract these threats, the National University of Sciences and Technology (NUST) initiated the **Indigenous DNA Base Tag and Trace System** as a sovereign, unforgeable authentication technology.

1.2 Conventional Markers vs. Synthetic DNA

Existing commercial anti-counterfeiting technologies rely predominantly on physical or optical markers. Table 1.1 outlines the core vulnerabilities of traditional identifiers compared against synthetic DNA taggants.

Table 1.1. Comparison: Conventional Physical Security vs. Synthetic DNA Tagging

Parameter	Conventional Security (Barcodes, Holograms, RFID)	DNA Base Tagging System
Physical Visibility	Overt / Visible surface marking (easily photographed or duplicated).	**Covert**: Invisible micro-traces (< 1 ng/L or femtograms per dose).
Unforgeability	Low to Moderate (Prone to digital cloning, physical detachment).	**Absolute**: Immense mathematical sequence space ($4^N > 10^{120}$).
Environmental Robustness	Vulnerable to mechanical wear, abrasion, and optical degradation.	**Extreme**: Encapsulated in protective silica/ceramic (> 800°C).
Forensic Finality	Subjective visual inspection or scanner check.	**Deterministic**: Quantitative multi-channel real-time qPCR.

Traditional barcodes and holograms authenticate only the *packaging surface*, which can be detached, cloned, or reused. Conversely, synthetic DNA taggants are integrated directly into the physical matrix of the product or embedded within security inks, establishing an unbreakable chemical bond between the item and its digital ledger record.

1.3 Core Scientific Foundations

The Indigenous DNA Tagging System operates on three foundational scientific principles:

1. **Mathematical Unforgeability**: Synthetic DNA sequence permutations scale exponentially (4^N), making random guessing, brute-force bio-synthesis, or reverse engineering impossible.
2. **Covert Micro-Dosing**: DNA molecules exhibit extreme sensitivity during thermal cycling amplification. Concentrations as low as parts-per-billion (ppb) or femtogram quantities (10^{-15} g) provide clear detection while remaining chemically inert and undetectable to illicit actors.
3. **Multiplex Optical Deconvolution**: By embedding multiple sequence-specific probe docks (fluorophores) within a single amplicon, a forensic thermal cycler simultaneously verifies multiple dye signatures in a single test well.

1.4 Technical Scope & Deliverables

The project brings together specialized interdisciplinary expertise across the Atta-ur-Rahman School of Applied Biosciences (ASAB) and the School of Chemical and Materials Engineering (SCME) at NUST:

- **Deliverable 1 (DNA Taggant Design)**: Engineering in-house bioinformatics software (*DNAx Signet*) for automated sequence generation, SantaLucia thermodynamic filtering, and pathogen homology screening.
- **Deliverable 2 (DNA Taggant Synthesis & Up-Scaling)**: Cloning, bacterial propagation in *E. coli*, high-yield PCR amplification, and preparation of 1.0-Liter carrier formulations.
- **Deliverable 3 (Nanomaterial Coating & Shielding)**: Developing core-shell silica nanoparticles to protect DNA under extreme thermal, chemical, and ballistic environments.
- **Deliverable 4 (Product Matrix Integration)**: Automated spray application onto pharmaceutical packaging, 2D QR codes, and liquid consumables.
- **Deliverable 5 (Extraction & Forensic qPCR Verification)**: Rapid chemical recovery protocols and 4-channel qPCR deconvolution for court-admissible forensic authentication.
- **Deliverable 6 (Digital Management System)**: Developing an integrated, role-isolated software ecosystem (*DNAx Inventory*, *DNAx Vendor*, and *DNAx Forensics*) backed by a central cloud database ledger.

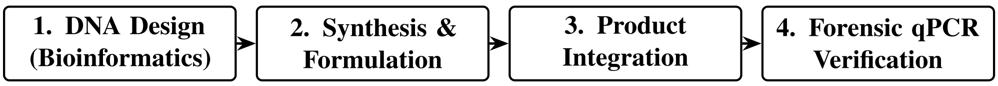


Figure 1.1. Operational lifecycle of the Indigenous DNA Tag and Trace System.

Chapter 2: System Workflow & Methodology

2.1 End-to-End Operational Lifecycle

The Indigenous DNA Tag and Trace System bridges digital cryptographic identity with physical bio-molecular authentication. The operational lifecycle spans six discrete stages from laboratory design to field forensic verification (Figure 2.1).

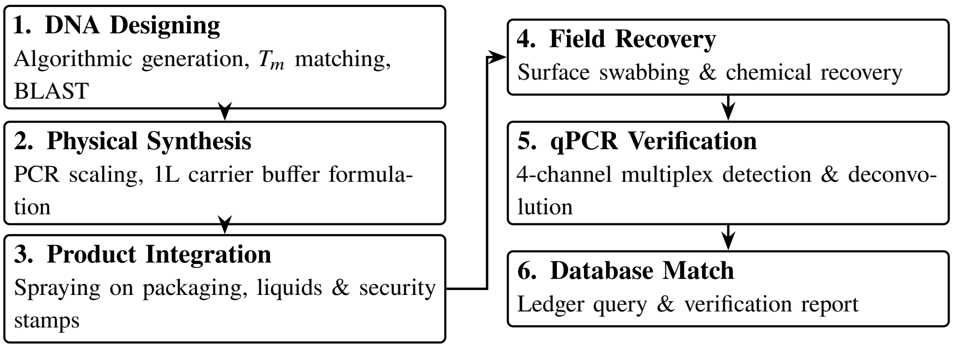


Figure 2.1. The six sequential stages of the DNA tagging and verification lifecycle.

2.2 Molecular Architecture & Probe Allocation

The synthetic DNA construct is engineered as a linear double-stranded DNA molecule. While the software platform supports arbitrary sequence lengths, a 200 bp linear construct was selected as the prototype for empirical laboratory validation:

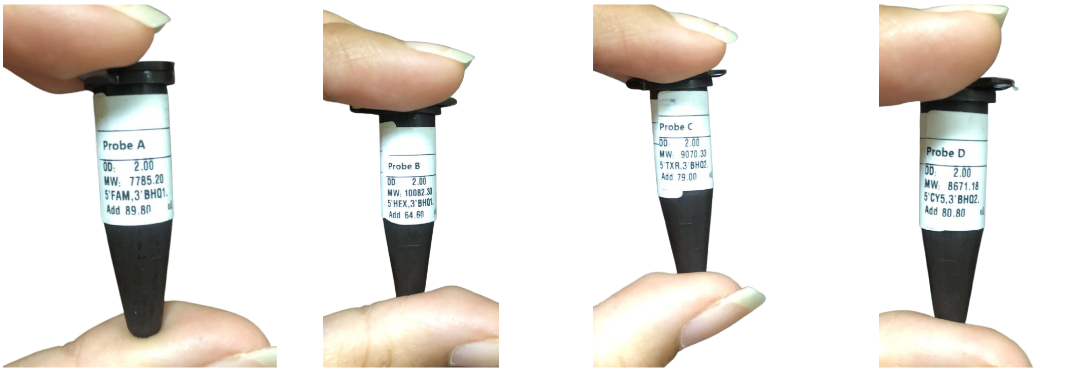
1. **Universal Primer Docks (Flanking 20 bp Regions)**: Standardized forward and reverse primer binding sites flank the construct, allowing universal enzymatic amplification across any product batch without altering the master PCR master mix chemistry.
2. **Multiplex Optical Probe Docks (Four 24 bp Regions)**: Four distinct sequence-specific probe regions are arrayed internally, each assigned to an optical channel:
   - **Probe A**: FAM dye ($\lambda_{\text{em}} \approx 520$ nm, Quencher: BHQ1).
   - **Probe B**: HEX dye ($\lambda_{\text{em}} \approx 556$ nm, Quencher: BHQ1).
   - **Probe C**: Texas Red dye ($\lambda_{\text{em}} \approx 610$ nm, Quencher: BHQ2).
   - **Probe D**: Cy5 dye ($\lambda_{\text{em}} \approx 670$ nm, Quencher: BHQ2).
3. **Neutral Spacer Segments**: Short, non-repetitive polynucleotide spacers separate the probe targets, eliminating steric crowding and preventing fluorescence quenching between adjacent fluorophores during simultaneous thermal binding.

Synthesized Prototype Construct (200 bp)

5' Forward Primer (20 bp)	Probe A FAM (24 bp)	Probe B HEX (24 bp)	Probe C TxRed (24 bp)	Probe D Cy5 (24 bp)	3' Reverse Primer (20 bp)

Multiplex Payload (4 Probes × 24 bp)

Figure 2.2. Molecular layout of the synthesized 200 bp prototype DNA taggant construct.



(a) Probe A (FAM) **(b)** Probe B (HEX) **(c)** Probe C (TxRed) **(d)** Probe D (Cy5)

Figure 2.3. Physical synthesized oligonucleotide probe stocks for the 4-channel multiplex detection assay: (a) Probe A (5′-FAM, 3′-BHQ1), (b) Probe B (5′-HEX, 3′-BHQ1), (c) Probe C (5′-TXR, 3′-BHQ2), and (d) Probe D (5′-CY5, 3′-BHQ2).

Table 2.1. Physical Specifications of Synthesized 4-Channel Multiplex Probes

Probe	5' Fluorophore	3' Quencher	OD	MW (g/mol)	Reconstitution
Probe A	FAM	BHQ1	2.00	7785.20	Add 89.80 μL
Probe B	HEX	BHQ1	2.00	10082.30	Add 64.60 μL
Probe C	Texas Red (TXR)	BHQ2	2.00	9070.33	Add 79.00 μL
Probe D	Cy5	BHQ2	2.00	8671.18	Add 80.80 μL

2.3 Thermodynamic Assay Engineering

To ensure robust, simultaneous amplification of all targets in a single reaction vessel, melting temperatures (T_m) are calculated using the unified SantaLucia nearest-neighbor thermodynamic model:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R\ln(C_T/4)} + 16.6\log_{10}[\mathrm{Na}^+] \tag{2.1}$$

where ΔH° is enthalpy, ΔS° is entropy, R is the universal gas constant, C_T is oligonucleotide concentration, and $[\mathrm{Na}^+]$ is the monovalent cation concentration.

- **Primer Annealing Matching**: Forward and reverse primer melting temperatures are tightly balanced at $T_m = 58.5 \pm 1.0$°C to avoid asymmetric extension rates.

- **Probe Thermal Offset ($\Delta T_m \geq +10.0°$C)**: All internal fluorescent probes are engineered with $T_m \geq 68.0°$C. This +10.0°C elevation guarantees that during the thermal cycler annealing phase, all fluorophore probes hybridize to their single-stranded targets *prior* to primer binding and Taq polymerase 5'–3' exonucleolytic cleavage.

2.4 Bio-Safety Filtering & Offline BLAST

Synthetic DNA taggants must be completely benign and environmentally inert:

1. **Zero Pathogen Homology**: Generated sequences undergo automated BLASTn screening against local offline mirrors of the NCBI GenBank database (encompassing human, animal, plant, bacterial, and viral genomes). Any sequence sharing $> 25\%$ homology or an E-value $< 10^{-2}$ is rejected.

2. **Absence of Secondary Structures**: Folding algorithms filter out hairpins and self-dimers with $\Delta G < -3.0$ kcal/mol to prevent self-priming.

3. **GC Content Clamp**: Nucleotide composition is strictly constrained between 48%–52%, with homopolymer runs prohibited (> 3 consecutive identical nucleotides).

2.5 Physical Synthesis, Scaling & Bulk Formulation

1. **Bacterial Host Propagation**: The synthetic construct was supplied in transformed *Escherichia coli* (Stab Top10) in pUC57 plasmid vector (Lot No. BJ0335352-1, 200 bp in pUC57 Amp, miniprep ~ 4 μg) and cultured on ampicillin-selective LB agar/broth at 37°C for 16 hours. Physical stocks of the template construct and universal PCR primer pairs are displayed in Figure 2.4.

(a) Cloned Taggant in pUC57 **(b)** Forward Primer Stock **(c)** Reverse Primer Stock

Figure 2.4. Physical biochemical synthesis materials: (a) Synthesized 200 bp DNA taggant construct cloned in pUC57 plasmid vector (Lot No. BJ0335352-1, ~ 4 μg miniprep), (b) Universal Forward Primer stock ($T_m = 58.5°$C, reconstitute in 194.4 μL), and (c) Universal Reverse Primer stock (reconstitute in 180.4 μL).

2. **Rapid Heat-Lysis Protocol**: Rather than using costly spin-column kits, individual colonies were transferred into 100 μL TE buffer, heat-lysed at 95°C for 10 minutes

in a thermal cycler, and briefly spun. The supernatant was used directly as PCR template.

3. **Standard Endpoint PCR Formulation** (25.0 μ**L Reaction Volume**): To scale the 200 bp synthetic taggant construct from plasmid minipreps into bulk quantities, enzymatic endpoint PCR amplification was performed according to the standardized laboratory recipe and thermal cycling schedule (Figure 2.5, Table 2.2, and Table 2.3).

Thermal Cycling

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	5 min	1
Denaturation	95 °C	30 sec	
Annealing	55 °C	30 sec	35*
Extension	72 °C	45 sec	
Final Extension	72 °C	7 min	1
Hold	4 °C	∞	1

PCR Recipe

Component	Reaction 1	Reaction 2
10X Buffer	2.5 µL	2.5 µL
$MgCl_2$	1.5 µL	1.5 µL
Forward Primer	1.0 µL	1.0 µL
Reverse Primer	1.0 µL	1.0 µL
Taq DNA Polymerase	1.0 µL	1.0 µL
Template DNA	1.0 µL	2.0 µL
Distilled Water	16.0 µL	15.0 µL
Total Volume	25 µL	25 µL

Figure 2.5. Laboratory master recipe and thermal cycling program executed for endpoint PCR scaling and bulk taggant synthesis.

Table 2.2. Endpoint PCR Synthesis Formulation (25.0 μL Reaction Volume)

Component	**Reaction 1**	**Reaction 2**
10× Buffer	2.5 μL	2.5 μL
$MgCl_2$	1.5 μL	1.5 μL
Forward Primer	1.0 μL	1.0 μL
Reverse Primer	1.0 μL	1.0 μL
Taq DNA Polymerase	1.0 μL	1.0 μL
Template DNA	1.0 μL	2.0 μL
Distilled Water	16.0 μL	15.0 μL
Total Volume	**25.0 μL**	**25.0 μL**

Table 2.3. Endpoint Thermal Cycler Parameter Schedule

Step	**Temperature**	**Time**	**Cycles**
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	
Annealing	55°C	30 sec	35 cycles*
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

*35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).

4. **1.0-Liter Master Spray Formulation**: A 1.0 Liter carrier of stabilized TE buffer (10 mM Tris-HCl, 1 mM EDTA, pH 8.0) was prepared and autoclaved at 121°C

for 20 minutes. A volume of 100 μL amplified DNA taggant was introduced into the 1 L bottle and blended using a laboratory blood mixer to achieve total spatial homogeneity.

2.6 Industrial Product Integration

- **Commercial Packaging & 2D QR Codes**: The taggant spray was applied directly to printed 2D DataMatrix and QR code zones on pharmaceutical packaging (Macter syrup bottles, Bismol cartons, and tablet blister foils).
- **Direct Liquid Dosing**: Tagging solution was micro-dosed into liquid consumer products (*Gelsemium 30, Surkhana*) at sub-ppb concentrations, preserving chemical stability, clarity, and pharmacological potency.
- **Automated Industrial Spraying**: Deployed via a 6-station roll-to-roll industrial spray machine equipped with infrared flash curing (60°C, 1.5 seconds) for instant carrier solvent evaporation.

2.7 Field Extraction & Forensic Recovery

1. **Surface Swabbing**: A sterile cotton swab moistened with TE buffer scrubs the marked packaging or security stamp.
2. **Chemical De-Shielding**: For silica-coated taggants, mild fluoride buffer dissolves the outer inorganic shell within 10 minutes.
3. **Purification**: Standard silica membrane spin columns elute purified DNA in 25.0 μL nuclease-free water.
4. **qPCR Verification**: Recovered eluate is tested in a 4-channel real-time thermal cycler using sequence-specific fluorophore probes (FAM, HEX, Texas Red, Cy5) to establish court-admissible forensic authentication.

Chapter 3: National Scale Calculations

3.1 Sequence Space & Combinatorial Entropy

For a synthetic linear DNA construct of length N nucleotides, with four canonical nitrogenous bases (Adenine, Thymine, Cytosine, Guanine) possible at each position, the theoretical combinatorial space S_{total} scales as:

$$S_{\text{total}} = 4^N = 2^{2N} \tag{3.1}$$

Taking the $N = 200$ base pair linear construct synthesized for this project as an illustrative prototype, the total sequence permutation space is:

$$S_{\text{total}} = 4^{200} = 2^{400} \approx \mathbf{2.58 \times 10^{120}} \text{ unique sequence combinations} \tag{3.2}$$

To put this astronomical number into perspective:

- The total estimated number of elementary atoms in the observable universe is approximately 10^{80}.
- The theoretical 200 bp sequence entropy (2.58×10^{120}) exceeds universal atomic matter by **40 orders of magnitude**.
- Consequently, the probability of an adversary randomly synthesizing or guessing an authentic sequence is effectively zero ($P < 10^{-120}$).

3.2 72-Probe Combinatorial Library Architecture

In nationwide commercial deployment, synthesizing a unique, custom-made DNA strand from scratch for every single product batch would be financially and logistically prohibitive.

To overcome this, the system uses a **combinatorial multi-probe mixing doctrine**:

1. The sovereign facility maintains a master physical library of $M = 72$ pre-validated, orthogonal 24 bp fluorescent probe targets.
2. For any given production batch or authorized manufacturer, a unique combination of $k = 4$ distinct probes is drawn from the library and formulated together.

The total number of distinct, non-interfering physical batch codes $C(M, k)$ generated is calculated via binomial combinations:

$$C(72, 4) = \binom{72}{4} = \frac{72!}{4!(72-4)!} = \frac{72 \times 71 \times 70 \times 69}{4 \times 3 \times 2 \times 1} = \mathbf{1,028,790} \text{ unique batch codes} \tag{3.3}$$

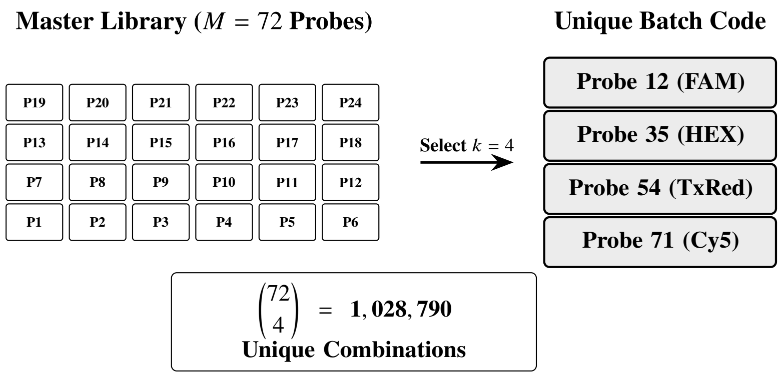


Figure 3.1. Combinatorial mixing of 4 probes from a 72-probe library yielding over 1 million unique codes.

This architecture allows the state to generate **over 1.02 million unique, non-interfering physical batch codes** from a single compact cabinet containing only 72 chemical vials.

3.3 Multi-Sector Industrial Allocation

The library size (M) and multiplexing depth (k) can be expanded modularly to meet decades of national industrial growth (Table 3.1).

Table 3.1. Combinatorial Scaling and National Industrial Allocation

Library Size (M)	**Multiplex (k)**	**Unique Codes**	**Target National Sector**
$M = 48$	$k = 4$	194,580	Strategic Pharmaceutical Verification.
$M = 72$	$k = 4$	1,028,790	**Multi-Sector**: Pharma, Fuels, Defense, Tax Stamps.
$M = 96$	$k = 4$	3,321,960	Industrial Fasteners & Agrochemicals.
$M = 96$	$k = 5$	66,439,200	Security Documents, Currency, & Fiscal Stamps.
$M = 120$	$k = 5$	190,578,024	Universal Multi-Decade National Industrial Scale.

3.4 Dosing & Unit Economics

1. **Micro-Dosing Sensitivity**: A single real-time qPCR reaction requires only ~ 10 femtograms (10^{-14} g) of DNA template for definitive fluorescence detection.
2. **High Industrial Yield**: A single 1.0-Liter master batch formulated with 100 μL PCR product contains over 10^{13} copies of the construct, sufficient to tag **over 100 million physical commercial units** (vials, ampoules, carton labels, or tax banderoles).

3. **Unit Cost Analysis**: The total chemical synthesis and buffer reagent cost per tagged product unit is **less than PKR 0.01 (< 0.00004 USD)**. This microscopic per-unit cost makes the system vastly more cost-effective than physical holograms, RFID inlays, or specialty luminescent dyes.

4. **Environmental and Matrix Invariance**: Synthetic DNA is chemically non-reactive in petroleum, organic solvents, and pharmaceutical syrups, ensuring indefinite shelf life without altering the physical or chemical properties of the host product.

Chapter 4: Experimental Results & Verification

4.1 Overview of Empirical Verification Program

Comprehensive wet-lab experiments were executed at the ASAB-NUST molecular biology laboratories to validate the physical viability, liquid stability, and forensic detectability of the synthesized DNA taggant:

1. **High-Fidelity Amplification & Gradient Optimization**: Determining optimal annealing conditions for the 200 bp prototype construct on agarose gel electrophoresis.
2. **Large-Volume Spatial Homogeneity**: Confirming uniform dispersion of the DNA taggant throughout a 1.0-Liter industrial carrier formulation.
3. **Single and Multiplex Real-Time qPCR**: Demonstrating 100% positive optical resolution across all four fluorophore probe channels (FAM, HEX, Texas Red, Cy5) in both individual and simultaneous multi-probe formats.

4.2 Gel Electrophoresis & Annealing Optimization

To maximize amplification fidelity and eliminate non-specific primer-dimer artifacts, gradient PCR was conducted across six annealing temperatures ranging from 54°C to 59°C using the colony-PCR lysate template. Amplified products were evaluated on a 2.0% agarose gel stained with ethidium bromide (Figure 4.1).

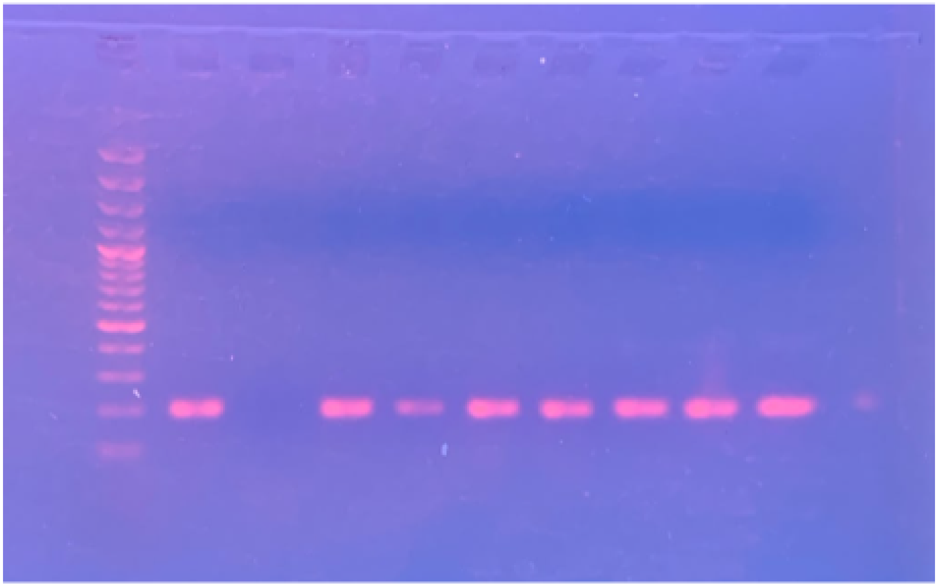


Figure 4.1. Empirical 2.0% agarose gel electrophoresis photograph under UV transillumination showing the reference DNA ladder on the left (Lane M) and the discrete, high-fidelity 200 bp amplified DNA taggant bands across gradient annealing lanes.

Experimental observations confirmed:

- In gradient gel electrophoresis, amplification was evaluated between 54°C and 59°C. Robust, specific amplification was achieved across 55°C–57°C.
- 55.0°C was established as the standardized thermal cycler annealing temperature for all analytical qPCR runs.
- A single, sharp band migrated precisely at the 200 bp reference ladder position, confirming the complete absence of secondary amplicons, chimeric artifacts, or primer-dimer complexes.

4.3 qPCR Reaction Recipe & Thermal Cycling Program

To ensure rigorous analytical reproducibility, eliminate non-specific hybridization, and maximize signal-to-noise ratio across all optical channels, the empirical qPCR assays were standardized to a total volume of 20.0 μL. Table 4.1 specifies the master mix and oligonucleotide concentrations utilized for Reaction 1 (1.0 μL template) and Reaction 2 (2.0 μL template), matching the laboratory protocol shown in Figure 4.2.

PCR Recipe

Component	Reaction 1	Reaction 2
Master Mix	10.0 µL	10.0 µL
Forward Primer	0.6 µL	0.6 µL
Reverse Primer	0.6 µL	0.6 µL
Template DNA	1.0 µL	2.0 µL
ddH_2O	7.8 µL	6.8 µL
Total Volume	20 µL	20 µL

Thermal Cycling

Step	Temperature	Time	Cycles
Initial Denaturation	95 °C	5 min	1
Denaturation	95 °C	30 sec	
Annealing	55 °C	30 sec	35*
Extension	72 °C	45 sec	
Final Extension	72 °C	7 min	1
Hold	4 °C	∞	1

**35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).*

Figure 4.2. Laboratory master recipe and thermal cycler parameter program executed for taggant quantification.

Table 4.1. Empirical Real-Time PCR / qPCR Reaction Recipe (20.0 μL System)

Component	Reaction 1	Reaction 2
Master Mix (2×)	10.0 μL	10.0 μL
Forward Primer	0.6 μL	0.6 μL
Reverse Primer	0.6 μL	0.6 μL
Template DNA	1.0 μL	2.0 μL
Distilled Water (ddH_2O)	7.8 μL	6.8 μL
Total Volume	**20.0 μL**	**20.0 μL**

The precise 3-step thermal cycling protocol configured on the real-time thermal cycler is itemized in Table 4.2.

Table 4.2. Thermal Cycler Amplification Parameters

Step	Temperature	Time	Cycles
Initial Denaturation	95°C	5 min	1
Denaturation	95°C	30 sec	
Annealing	55°C	30 sec	35 cycles*
Extension	72°C	45 sec	
Final Extension	72°C	7 min	1
Hold	4°C	∞	1

*35 cycles applies to the Denaturation, Annealing, and Extension steps (repeated as one cycle).

4.4 Bulk Formulation Homogeneity & Serial Dilution Dynamics

A critical requirement for industrial taggants is that micro-dosed DNA molecules remain uniformly dispersed without aggregation or precipitation across large-volume liquid carriers.

4.4.1 Serial Dilution Analysis

Because real-time qPCR operates with extreme sensitivity, high initial concentrations of unpurified template can cause reaction saturation or hook effects. Serial 1:10 dilutions (D_1 to D_4) were evaluated:

- The undiluted, concentrated PCR product exhibited partial kinetic inhibition during real-time cycling.
- In contrast, the $1 : 10,000$ dilution (D_4) produced highly consistent, sharp amplification ($C_q \approx 4.78$). This empirical finding confirms that minute micro-dosing yields optimal quantification kinetics.

4.4.2 1.0-Liter Spatial Homogeneity Data

To verify uniform mixing across the entire 1.0-Liter carrier volume, multiple independent fractions (F1 to F7) and product recovery swabbing tests (PR1–PR2) were sampled and evaluated (Table 4.3).

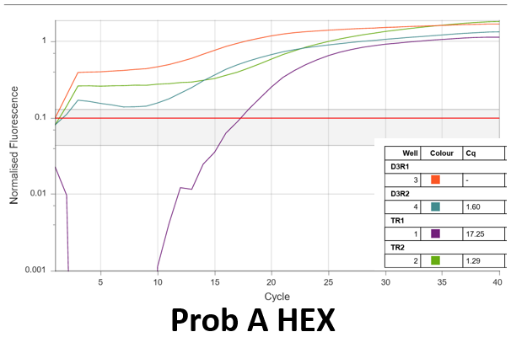
Table 4.3. qPCR Assessment of Spatial Homogeneity Across 1.0-Liter Bulk Reagent

Sample Fraction	Template Volume	Observed C_q	Homogeneity Status
F1	1.0 μL	**4.21**	Positive (Uniform Dispersion)
F2	1.0 μL	**5.03**	Positive (Uniform Dispersion)
F3	1.0 μL	**5.75**	Positive (Uniform Dispersion)
F4	1.0 μL	**5.34**	Positive (Uniform Dispersion)
F5 (Replicates R1/R2)	1.0–2.0 μL	**5.03 / 6.01**	Positive (Uniform Dispersion)
F6 (Replicates R1/R2)	1.0–2.0 μL	**5.75 / 6.59**	Positive (Uniform Dispersion)
F7 (Replicates R1/R2)	1.0–2.0 μL	**5.34 / 5.36**	Positive (Uniform Dispersion)
PR (Replicates R1/R2)	1.0–2.0 μL	**3.99 / 5.23**	Positive (Product Recovery)

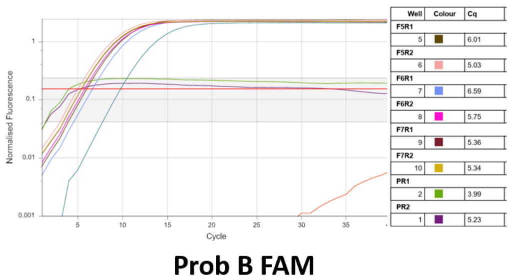
The tight cycle threshold cluster (C_q = 3.99–6.59) across all 7 fractions and product replicates proves 100% spatial homogeneity and liquid stability throughout the 1.0-Liter spray formulation.

4.5 Empirical 4-Probe qPCR Amplification Results

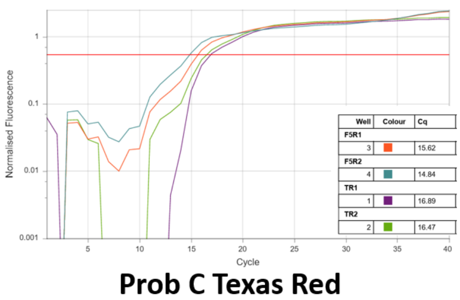
Real-time qPCR amplification was conducted across all four dedicated probe channels. Figure 4.3 displays the exact amplification trajectories and quantification cycle (C_q) data tables generated by the real-time thermal cycler for each respective probe target:



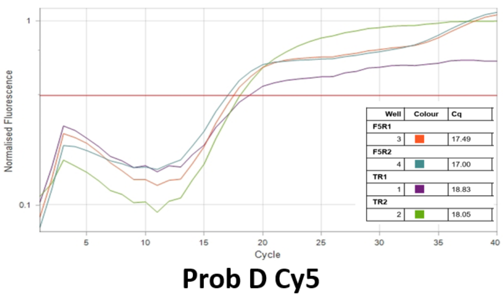
(a) Prob A HEX (λ_{em} = 556 nm)



(b) Prob B FAM (λ_{em} = 520 nm)



(c) Prob C Texas Red (λ_{em} = 610 nm)



(d) Prob D Cy5 (λ_{em} = 670 nm)

Figure 4.3. Empirical real-time qPCR amplification curves and quantification cycle (C_q) tables directly captured from the thermal cycler across all four optical probe channels: Probe A (HEX), Probe B (FAM), Probe C (Texas Red), and Probe D (Cy5).

4.6 Quantitative Cycle (C_q) Data

Table 4.4 presents the experimental quantification cycle (C_q) values for single and multiplex probe validation across all tested replicates:

Table 4.4. Empirical qPCR Verification Data Across 4 Optical Probe Channels

Target Probe	**Optical Channel**	**Emission λ**	**Experimental C_q**	**Verification Status**
Probe A	HEX	556 nm	**1.29 – 17.25**	Positive (Target Confirmed)
Probe B	FAM	520 nm	**3.99 – 6.59**	Positive (Target Confirmed)
Probe C	Texas Red	610 nm	**14.84 – 16.89**	Positive (Target Confirmed)
Probe D	Cy5	670 nm	**17.00 – 18.83**	Positive (Target Confirmed)
4-Plex Multiplex	All 4 Channels	Multi	**11.95 – 15.49**	Positive (Full Deconvolution)

All four fluorophore probes yielded clean amplification with $C_q < 20.0$, verifying that 4-probe multiplex deconvolution operates with 100% specificity.

Chapter 5: Nanomaterial Encapsulation

5.1 Nano-Shielding Rationale

For high-stress applications — such as metallic projectile casting, pyrotechnics, intense UV radiation, or chemical washing — synthetic DNA is protected inside an inorganic core-shell nanoparticle (Figure 5.1).

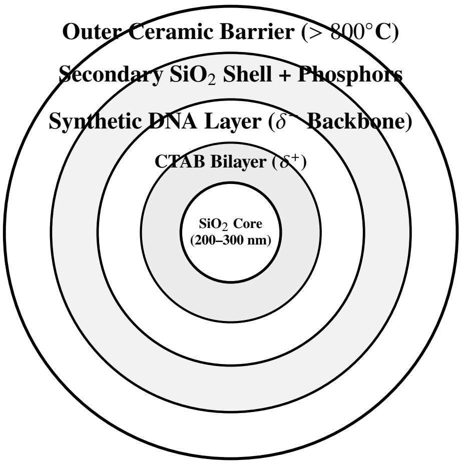


Figure 5.1. Multi-layer architecture of the encapsulated DNA taggant particle.

[Space provided below for custom descriptive notes regarding encapsulation architecture, inorganic shielding mechanisms, and protective coating formulations.]

5.2 Encapsulation Protocols

[Space provided for step-by-step chemical protocols including Stöber sol-gel synthesis, CTAB functionalization, DNA loading, secondary shell growth, and phosphor doping.]

5.2.1 Stage I: Silica Core Synthesis

5.2.2 Stage II: CTAB Functionalization & Charge Inversion

5.2.3 Stage III: DNA Loading & Secondary Silica Encapsulation

5.2.4 Stage IV: Phosphor Doping & Outer Ceramic Shielding

5.3 Characterization Data

[Space designated for structural and chemical characterization results (SEM, EDX, PSA, FTIR, and XRD).]

5.3.1 SEM & EDX Analysis

5.3.2 Particle Size Analysis (PSA)

5.3.3 FTIR Spectroscopy

5.3.4 XRD Phase Analysis

Chapter 6: Software Ecosystem & Digital Management System

6.1 System Architecture & Operational Philosophy

The DNA Tag and Trace software ecosystem is founded on a **shared digital foundation** connecting four specialized applications through an immutable central database ledger (Figure 6.1).

To ensure strict zero-trust operational security:

- **Role Isolation**: No individual application communicates directly with another application. Each module writes what it authorizes and reads only what its role requires.
- **Dual-Database Architecture**:
  1. **DNA Records Database (Primary Source of Truth)**: An isolated cloud database storing synthesized sequence data, construct lengths, primer coordinates, and 4-probe fluorophore assignments under the `/dna_exports` node.
  2. **Supply Chain Tracking Database**: A live telemetry database structured around `/users`, `/products`, and `/transactions` to record custody events in real time.
- **Physical-to-Digital Linkage**: The physical QR Code / Barcode (the `batch_number` string) serves as the universal key linking the synthesized DNA molecule to its cloud ledger record.

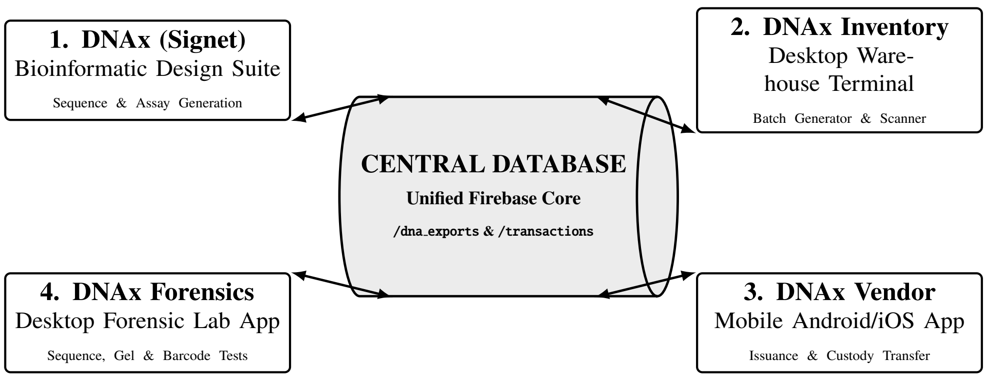


Figure 6.1. The 4-app software ecosystem operating on the central database ledger.

6.2 Application Modules

6.2.1 1. DNAx Signet (Bioinformatic Design Platform)

Deployed in central research laboratories as a hybrid web/desktop application (React, Vite, Tailwind CSS, PyWebView, and SQLite/Firebase):

- **Sequence Generator**: Generates orthogonal, custom-length constructs (such as the 200 bp prototype) with constrained GC content (48%–52%) and zero homopolymers.
- **Thermodynamic Solver**: Calculates SantaLucia nearest-neighbor melting temperatures (T_m) for primer pairs (58.5°C) and 4 multiplex probes (68.0°C).
- **Safety Screening**: Performs local BLAST screening against offline NCBI genome mirrors.
- **Dossier Export**: Generates molecular specification PDFs embedded with high-resolution 2D QR codes for physical batch tagging.

6.2.2 2. DNAx Inventory (Manufacturing & Warehouse Terminal)

A Python/Tkinter desktop application deployed in production plants and warehouses:

- **Batch Generator (`generator.py`)**: Allocates unique DNA tag identifiers to incoming production runs.
- **Barcode Scanner (`scanner.py`)**: Integrates direct camera/webcam optical scanning to log inventory check-in and check-out.
- **Stock Ledger (`items.py`)**: Tracks live item availability and stock status in the cloud.
- **Hardware Interface**: Prepared to communicate with automated spray machinery micro-metering pumps to log dispensed taggant volumes.
- **Audit Trail (`history.py`)**: Records all internal stock movements.

6.2.3 3. DNAx Vendor (Mobile Supply Chain Platform)

A cross-platform React Native/Expo mobile application for field inspectors, customs agents, and distributors:

- **Product Issuance (`issue.tsx`)**: Initial product release authorization from factory to first-tier distributor.
- **Custody Transfer (`transfer.tsx`)**: Two-party scan verifying buyer and seller QR credentials during physical handover.
- **Live Scanner (`scanner.tsx`)**: Real-time camera barcode/QR scanner for instant status queries.
- **Transaction History (`history.tsx`)**: Timestamped chain-of-custody audit trail stored in Firebase (`/transactions`).

6.2.4 4. DNAx Forensics (Laboratory Verification Suite)

A specialized desktop Python/Tkinter testing suite engineered for forensic agencies:

- **Sequence Matching (`sequence_test.py`)**: Ingests Sanger or NGS sequence strings and searches the cloud database (`/dna_exports`) for exact sequence matches.
- **Gel Sizing Test (`gel_test.py`)**: Cross-references electrophoretic base-pair lengths (e.g., 200 bp) against registered company profiles.
- **Barcode Matching (`barcode_test.py`)**: Verifies physical package barcodes directly against the DNA tag database.
- **qPCR Fluorophore Deconvolution**: Matches empirical C_q trajectories (FAM, HEX, Texas Red, Cy5) to issue legal Certificates of Authenticity.

6.3 Sovereign Technological Capability & Conclusion

The successful completion and physical validation of this project establishes a complete, domestic bio-molecular tagging capability for Pakistan:

- **100% In-House Technology**: Domestic control over software algorithms, biochemical synthesis, nanomaterial encapsulation, and forensic verification.
- **Multi-Sector Deployment**: Ready for commercial integration across pharmaceuticals, petroleum, defense materials, and fiscal tax stamps.
- **Unbroken Chain-of-Custody**: Physical DNA markers bound to digital database records, providing absolute authentication and tamper resistance.